University
Mohammed VI
Polytechnic

College of
Computing

Deliverable #: Title of Lab/Project here (e.g.: Conceptual Design ....)

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Team Name	UNO
Member 1	Daniel Tchonang
Member 2	Amine Sira
Member 3	Mohammed Oumane
Member 4	Zaid Zehari
Repository Link	`https://github.com/unocomputing/unodatabase.git`

1 Roles

- Daniel: Report
- Sira: Report
- Mohammed: Report
- Zaid: Report & determination of FDs

2 Challenges

No significant challenges were encountered during the project work

3 List of FDs

For each table, the primary key determines all non-key attributes. The functional dependencies are:

- `Person`: `pid` → `firstName, lastName, email, affiliation, startDate, endDate`
- `Student`: `pid` → `program`
- `Employee`: `pid` → `phone, office, supervisorPid`
- `Academic`: no non-trivial FD.
- `NonAcademic`: no non-trivial FD.
- `Faculty`: `pid` → `position`
- `Administrative`: `pid` → `position`
- `Technical`: `pid` → `position`
- `Advises`: `(academicPid, studentPid)` → all attributes
- `Laboratory`: `labId` → `name, building, roomNumber, discipline, supervisorPid`
- `Attached`: `(pid, labId)` → all attributes
- `ResearchProject`: `code` → `title, startDate, endDate, status`
- `Participates`: `(pid, projectCode)` → `role`
- `Budget`: `budgetLine` → `amountGranted, amountDisbursed, startDate, endDate, managerPid`
- `FundsPrj`: `(budgetLine, projectCode)` → all attributes
- `FundsLab`: `(budgetLine, labId)` → all attributes
- `EquipmentModel`: `modelId` → `commercialName, manufacturer, category, requiredEnvironm trainingMandatory`

- `EquipmentUnit`: `serialNo` → `acquisitionDate, purchaseCost, status, portable, modelId, labId`
- `Certification`: `code` → `title, issuingAuthority, validityPeriod, safetyLevel`
- `Requires`: `(modelId, certCode)` → all attributes
- `Holds`: `(pid, certCode)` → `issueDate, expirationDate, grade`
- `Reservation`: `resId` → `submissionTS, plannedStart, plannedEnd, purpose, status, madeByPid, projectCode, approverPid`
- `Reserves`: `(resId, serialNo)` → all attributes
- `Maintenance`: `(serialNo, startTS)` → `endTS, type, description, cost, outcome, technicalPid`
- `CalibrationRecord`: `(serialNo, calibDate)` → `calibrationType, result, nextDueDate, remarks`
- `Consumable`: `consId` → `name, unitOfMeasure, hazardLevel, reorderThreshold`
- `Supplier`: `suppId` → `name, contactEmail, phone`
- `Supplies`: `(suppId, consId, labId)` → `unitPrice`
- `Stocks`: `(labId, consId)` → `quantityOnHand, lastRestockDate, storageCondition, technicalPid, monitoringSince`
- `Consumes`: `(resId, labId, consId)` → `quantityUsed`

4 Justification of Functional Dependencies

Some tables, such as `Academic`, `NonAcademic`, `Advises`, `Attached`, `FundsPrj`, `FundsLab`, `Requires`, and `Reserves`, have no non-trivial functional dependencies beyond those implied by their primary keys.

This is because these tables mainly represent relationships or subtypes and do not contain attributes whose values are determined by something other than the complete primary key. Therefore, no additional FD can be derived from the requirements or business rules for these relations.

5 Assumptions

We assumed that it was not necessary to explicitly list all superkey-based functional dependencies, since they can be derived from the functional dependencies determined by the primary keys. We therefore list the primary-key FDs and the additional non-trivial FDs relevant to the RLMS requirements.